Wide-and-Native Co-Packaged Optics: Backside-Emitting µVCSEL Arrays and Multicore Fiber

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LightXcelerate Inc.

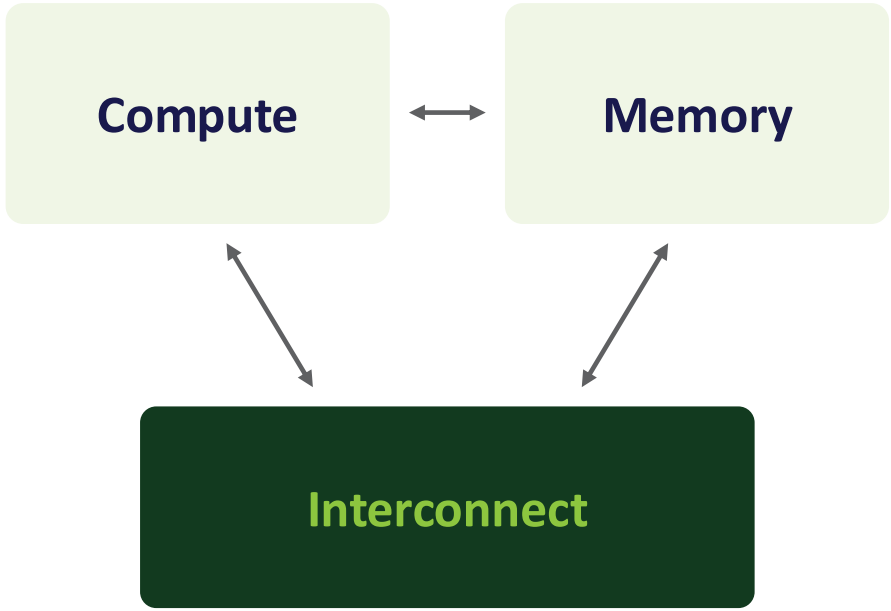
weipeng@lightxcelerate.com

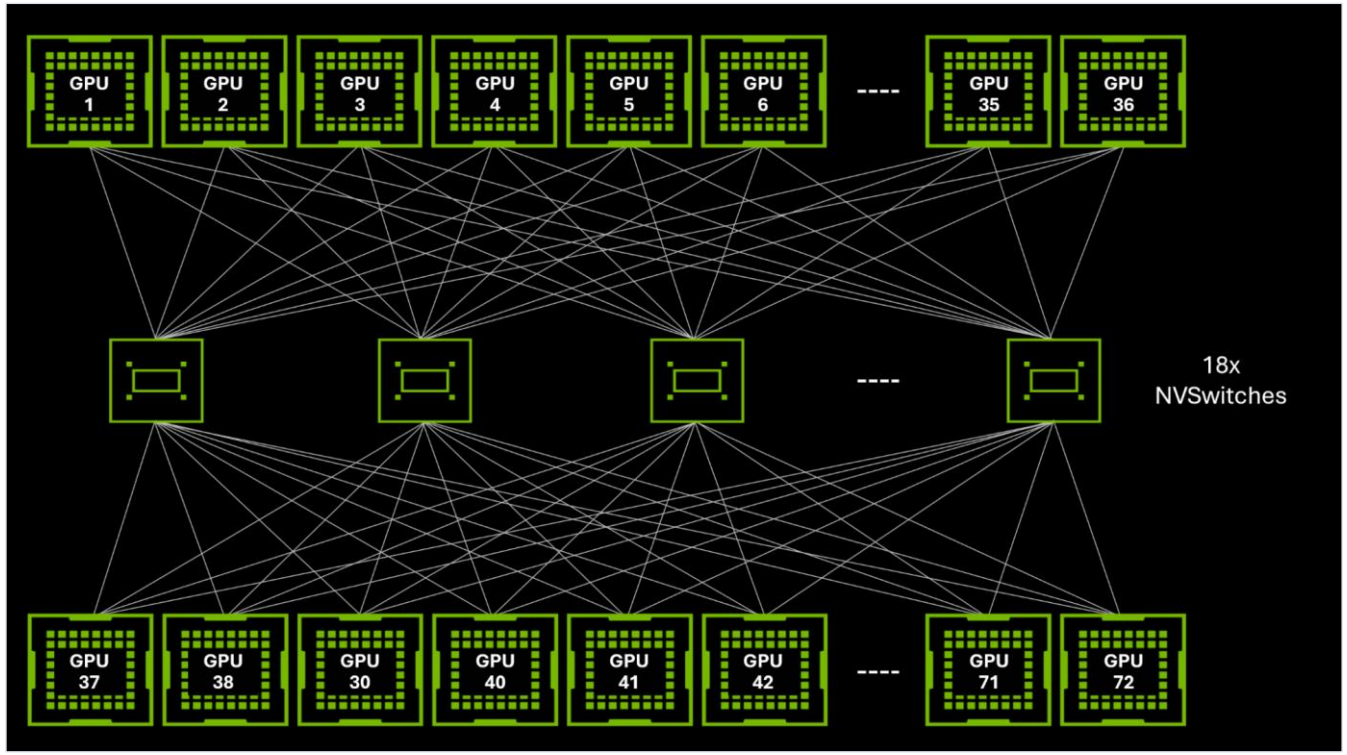


OCP FUTURE TECHNOLOGIES SYMPOSIUM

October 14, 2026
San Jose, California

The I/O Wall Is Stifling AI Performance





GB200 NVL72 scale-up domain: 72 GPUs connected through 18 NVSwitches. Image: NVIDIA Technical Blog.

Scaling AI data centers requires **far more aggressive bandwidth scaling** than traditional pluggable optics was designed to support.

Strategic Capital Is Moving to Optics

More than $15B in announced strategic commitments since late 2025.

$6B

NVIDIA optics ecosystem investments

$2B Lumentum

$2B Coherent

$2B Marvell

Announced March 2026.

$3.25B

Marvell acquires Celestial AI

Completed Feb 2026.

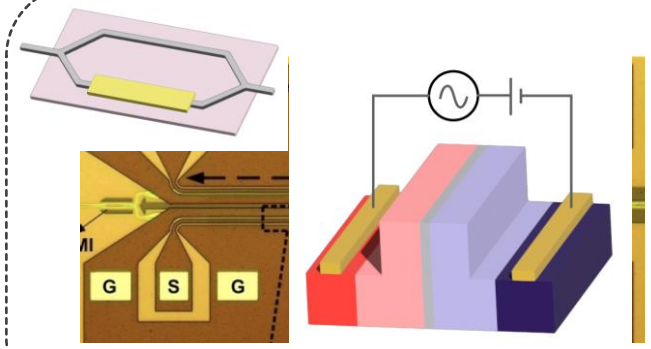
Up to $6B

Meta–Corning optical agreement

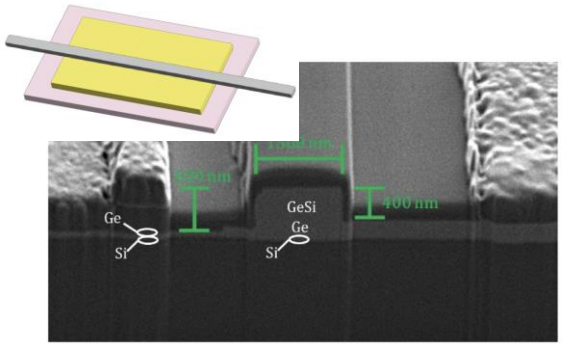
Multiyear supply of fiber-optic connectivity for Meta's AI data centers. A commercial supply agreement, not an equity investment.

µVCSEL: The Solution of Choice

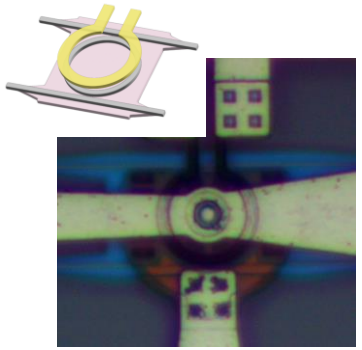
Silicon Photonics



MZM



EAM



MRM

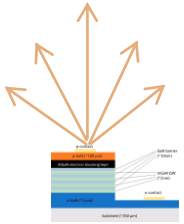
+

External Laser



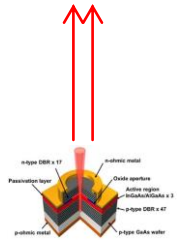
~ 10,000mm

Direct Modulation
(No External Parts)



~ 50 um

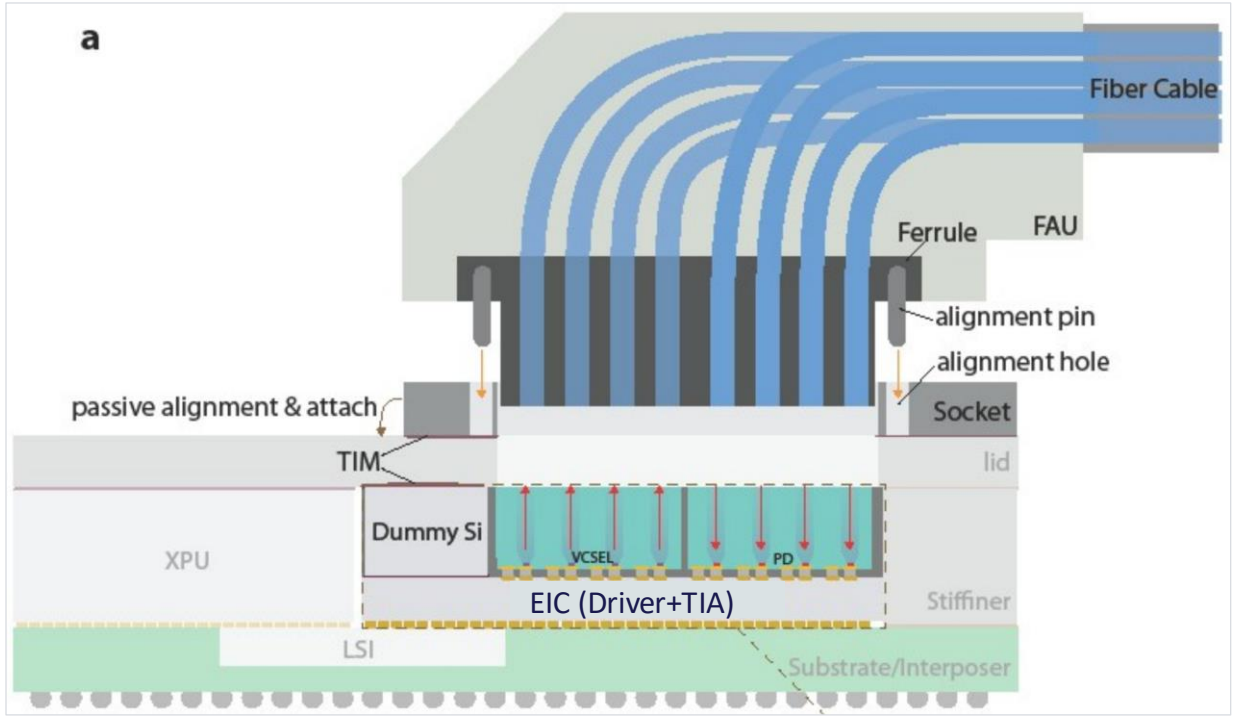
Micro-LED



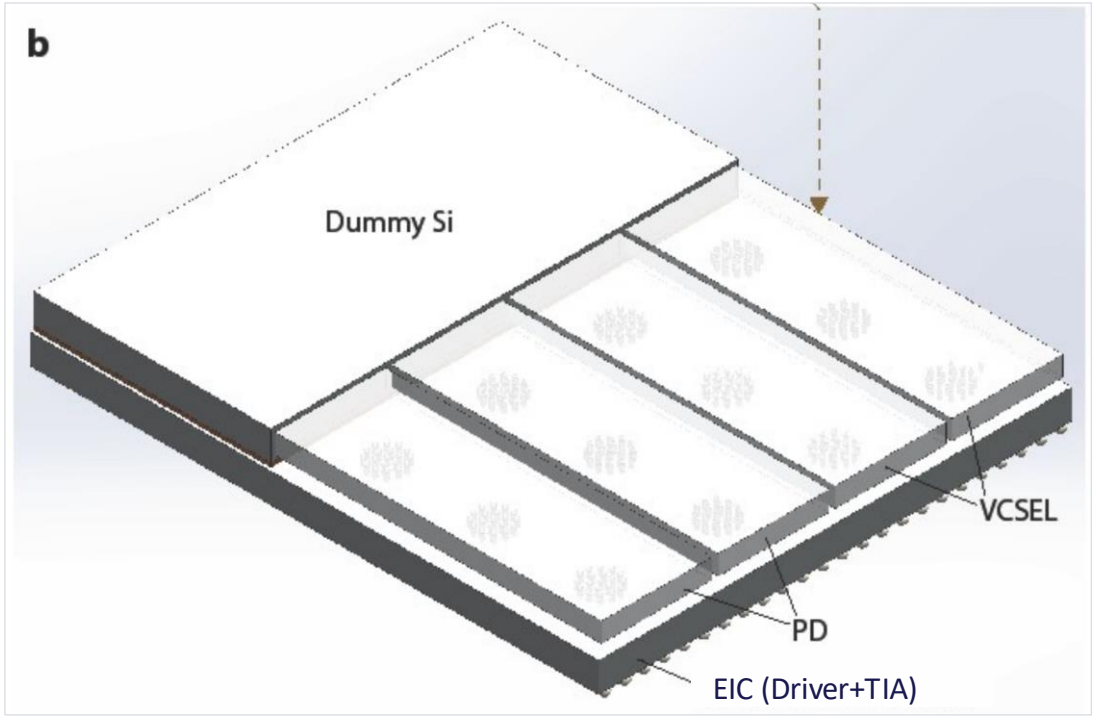
~ 50 um

Micro-VCSEL

Optical Engine Architecture



(a) Cross-section: engine beside the XPU on a shared substrate/interposer.



(b) Chiplet: VCSEL and PD dies tiled on the EIC.

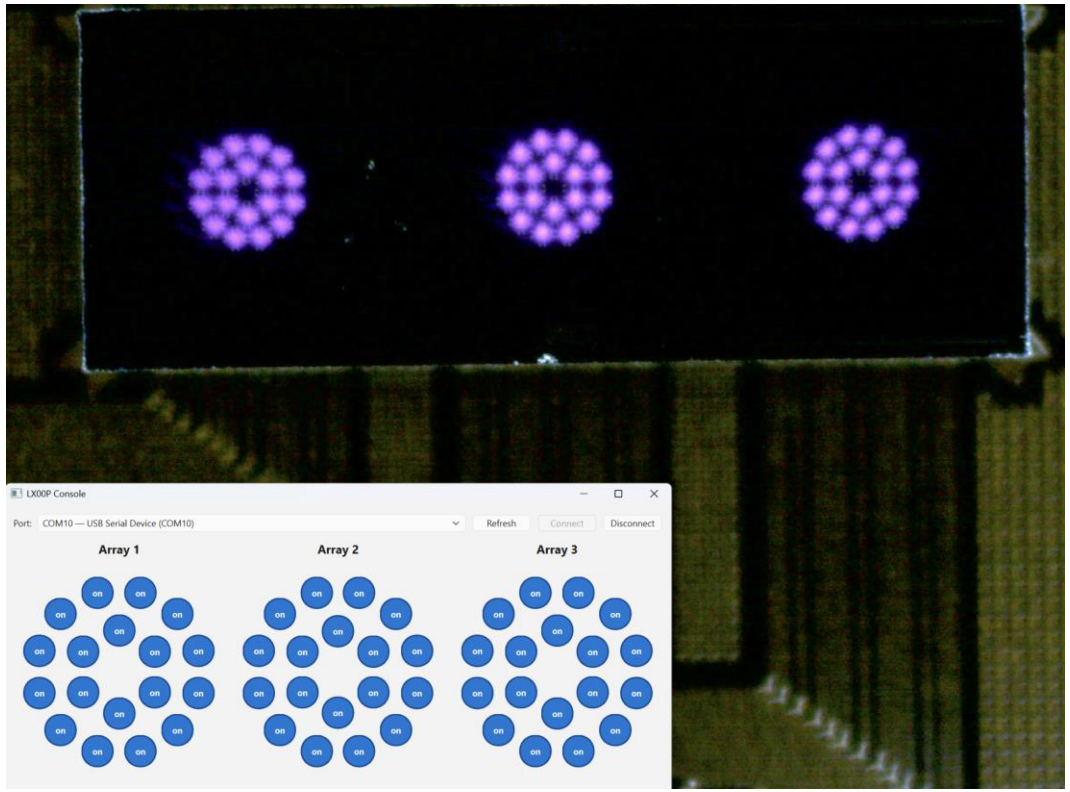
1. **Passive fiber attach**
   FAU pins drop into a socket on the lid.

2. **Backside optics**
   BSE VCSELs + BSI PDs flip-chipped face-down.

3. **EIC underneath**
   Driver + TIA; µm-scale Cu-pillar paths.

4. **Thermal path**
   Dummy Si + TIM carry heat into the lid.

LightX-1: A Modular µVCSEL Optical-Engine Block



Three 19-channel µVCSEL subarrays under drive, with per-channel addressing (inset).

57 lanes
57 Tx + 57 Rx

1.5 × 0.6 mm
engine footprint

32 Gb/s
per lane today, roadmap to 64G

3.2 Tb/s
per engine

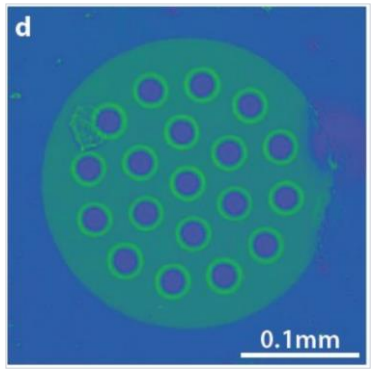
< 1.5 pJ/bit
full link

50 m
reach

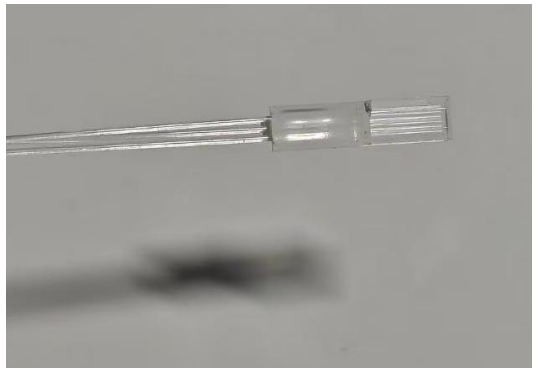
µVCSELs ~25× smaller than conventional VCSELs; roadmap to 100 and 200 Gb/s per lane.

µVCSEL Array × Multicore Fiber

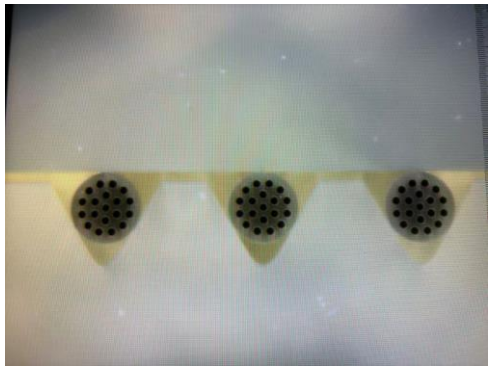
Partnership with YOFC Smart Link Better Life.



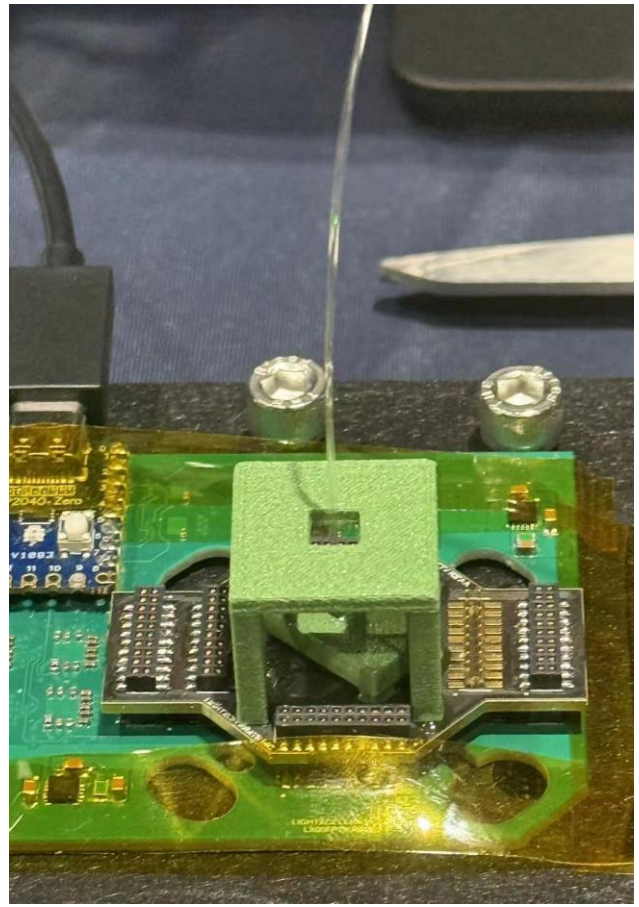
19-core MCF.



Fiber array



End surface of fiber array

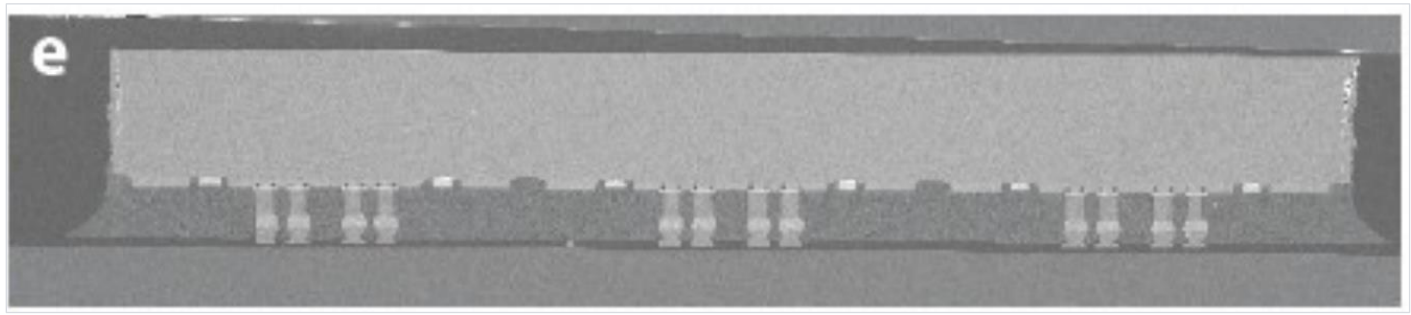


Test vehicle with MCF pigtail.

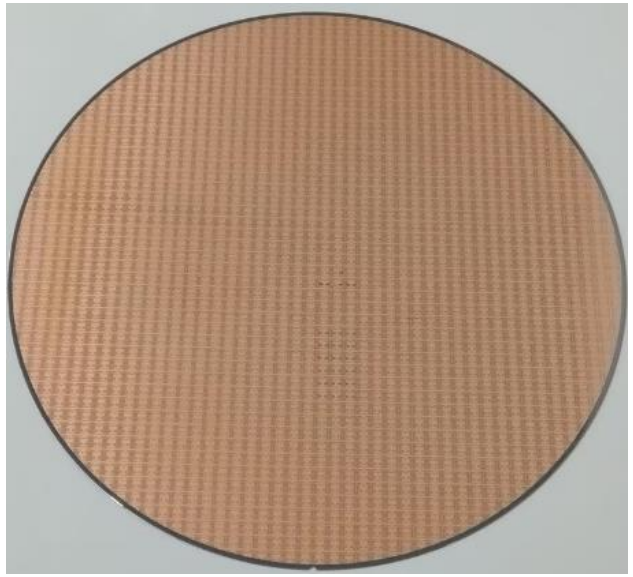
One MCF replaces a bundle of 19 fibers:

Lower cost per channel	**Smaller** cable	**Tighter** bends	**1:1** Pattern-matched coupling

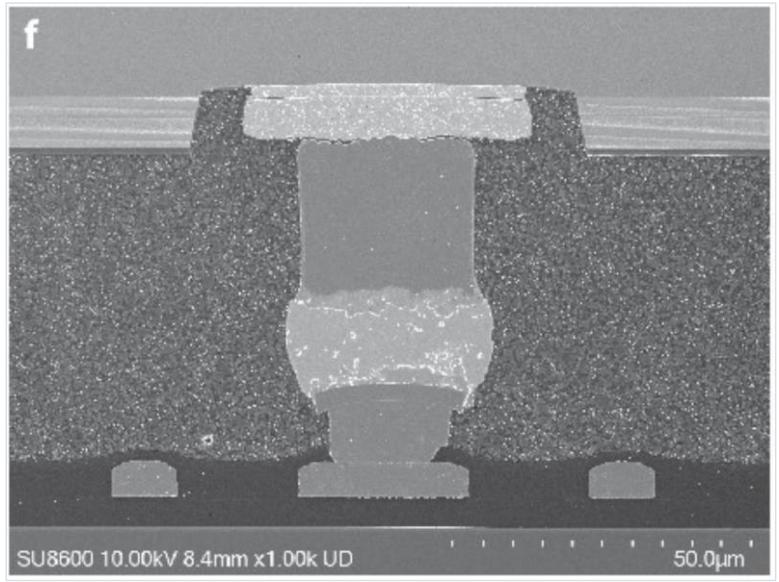
Flip-Chip Assembly with Cu Pillars



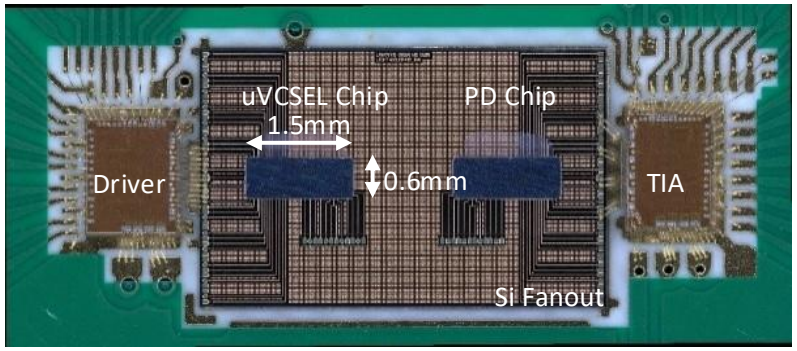
SEM cross-section of the flip-chip bonded VCSEL die.



12in Interposer wafer.



Magnified view of a bonded Cu pillar.

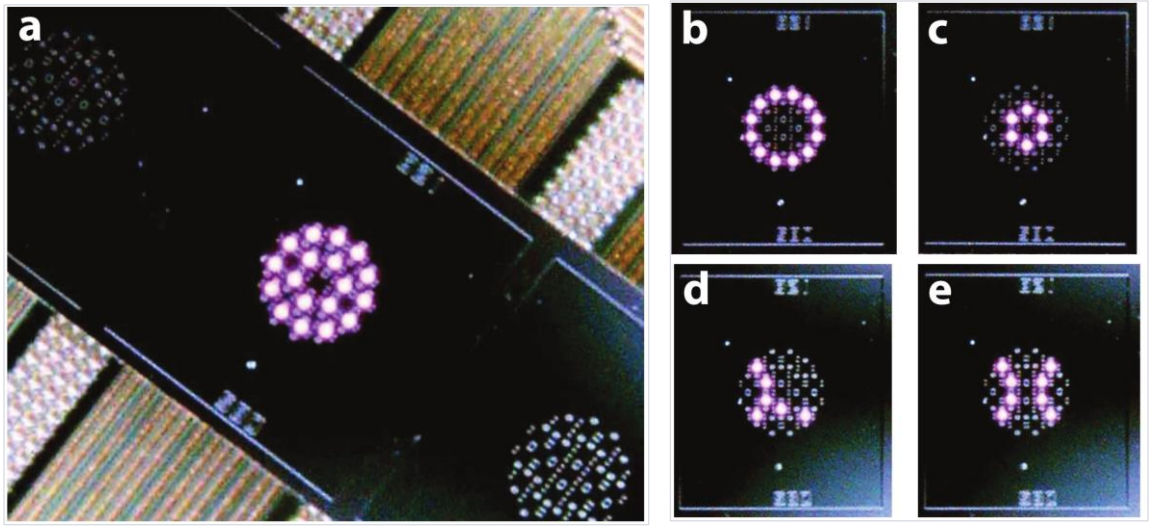


Fully packaged test bench

Built on OSAT-ready steps

- Cu-pillar flip-chip bonding of III-V dies
- Underfill for mechanical robustness
- TSVs route vertical I/O through the EIC
- Passive mechanical fiber alignment
- Conventional package integration

DC and Array Operation



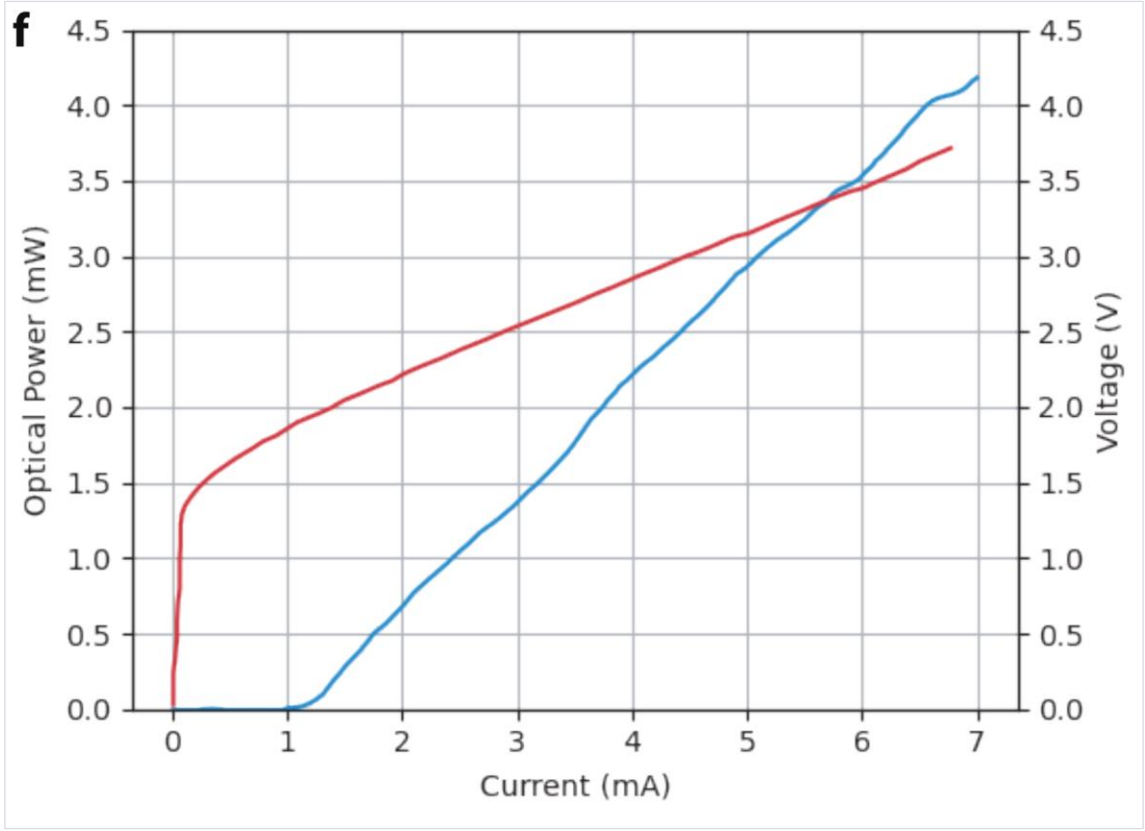
(a) Electrically driven BSE VCSEL array; (b–e) emission patterns under different drive conditions.

≈1.1 mA
lasing threshold

≈3.5 mW
output @ 6 mA

≈0.7 W/A
slope efficiency

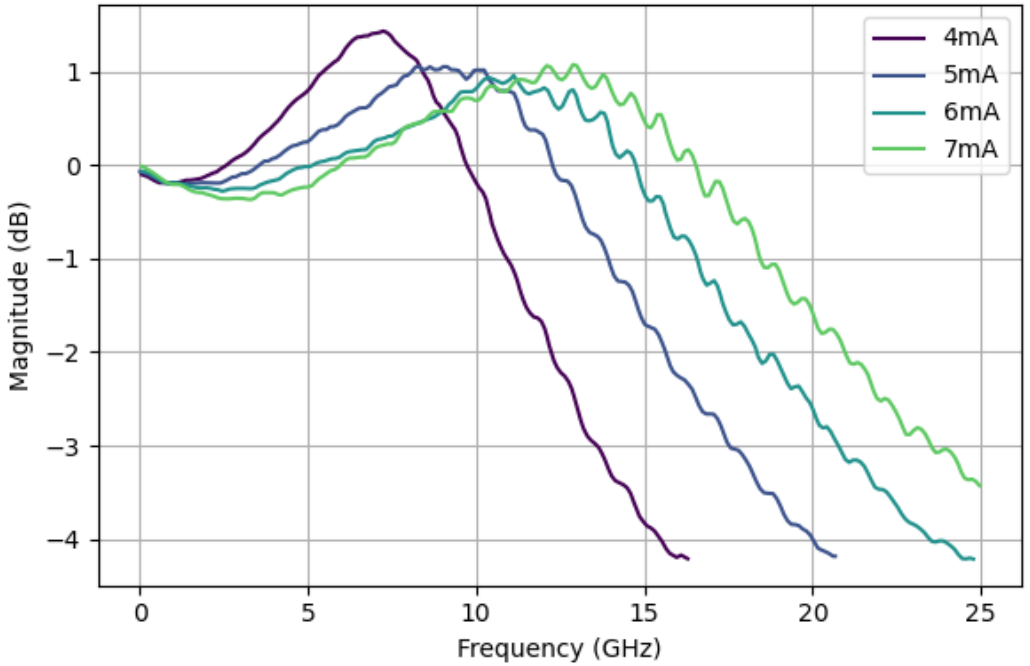
L–I–V characteristics



Optical power (blue, left) and voltage (red, right) vs. drive current.

High-Speed Operation

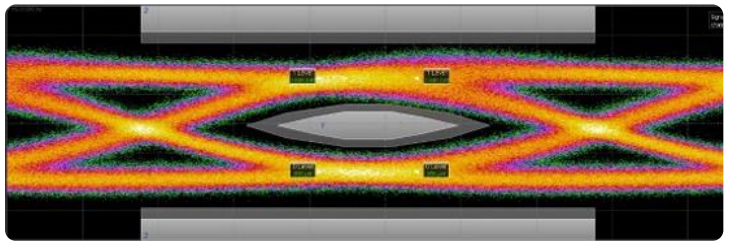
Small-signal response vs. bias



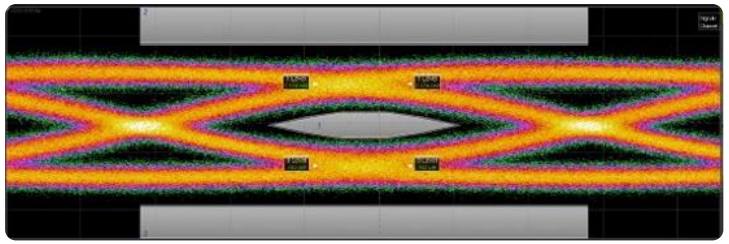
~20 GHz
S21 BW @ 7 mA

32 Gb/s
NRZ

Measured eye diagrams

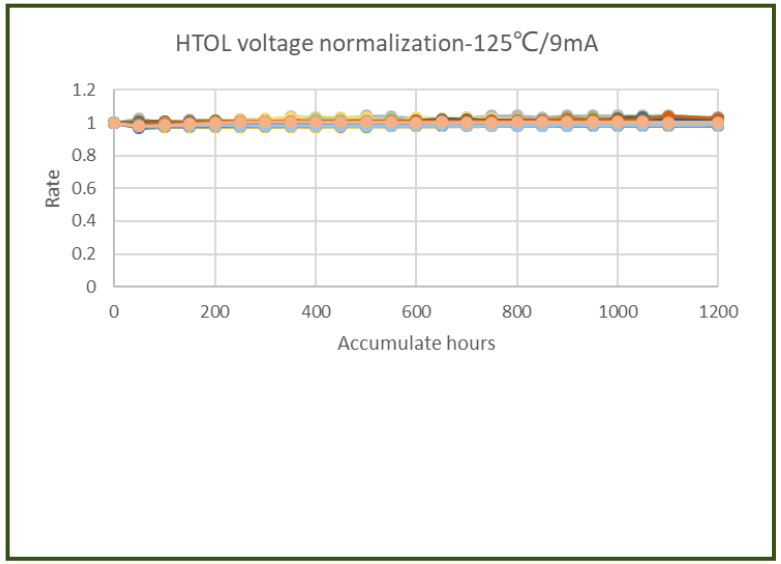


25 Gb/s · D7 · 6 mA · 0.7 Vpp
Mask margin 23.1% · ER 2.39 dB

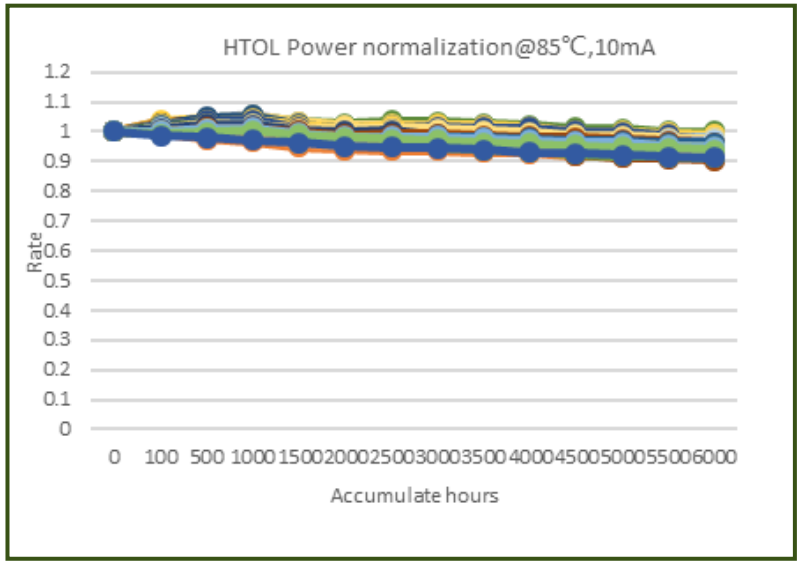


32 Gb/s · D7 · 7 mA · 0.7 Vpp
Mask margin 5.5% · ER 1.88 dB

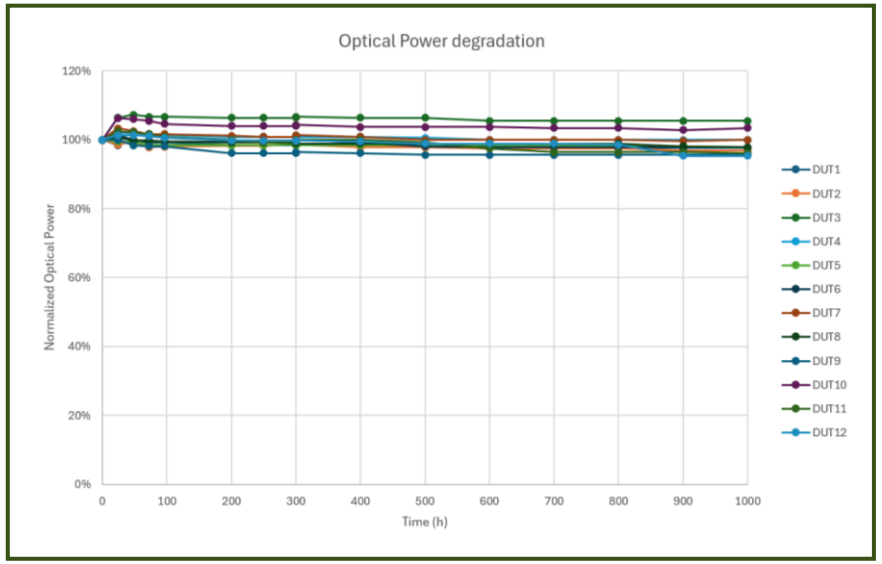
Reliability



HTOL 125 °C / 9 mA



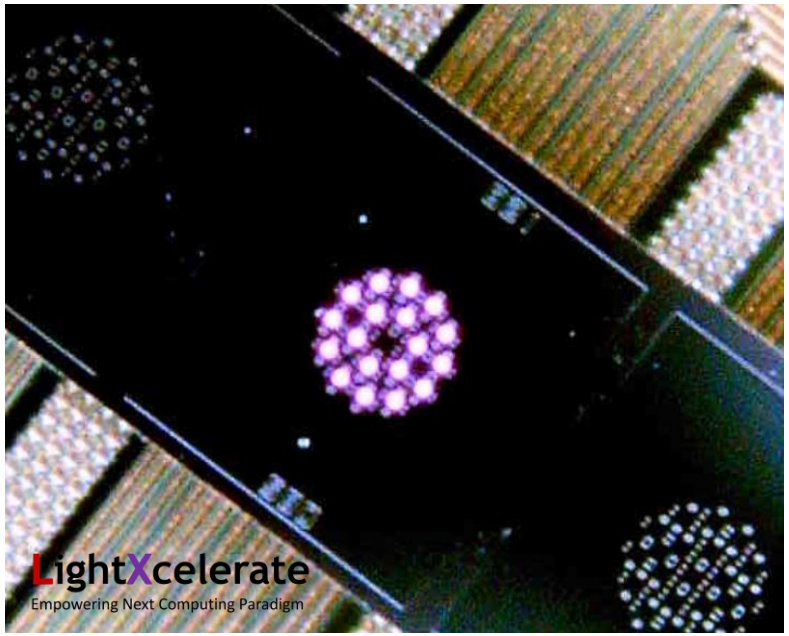
HTOL 85 °C / 10 mA: normalized power.



OA 3um, 6 mA, Tj 165C

An Exciting Time for Optical I/O

57 lanes	2.5D	MCF & FAU	< 1.5 pJ/bit	50 m
57 Tx + 57 Rx	packaging	optical coupling	Full link	reach



Join Us

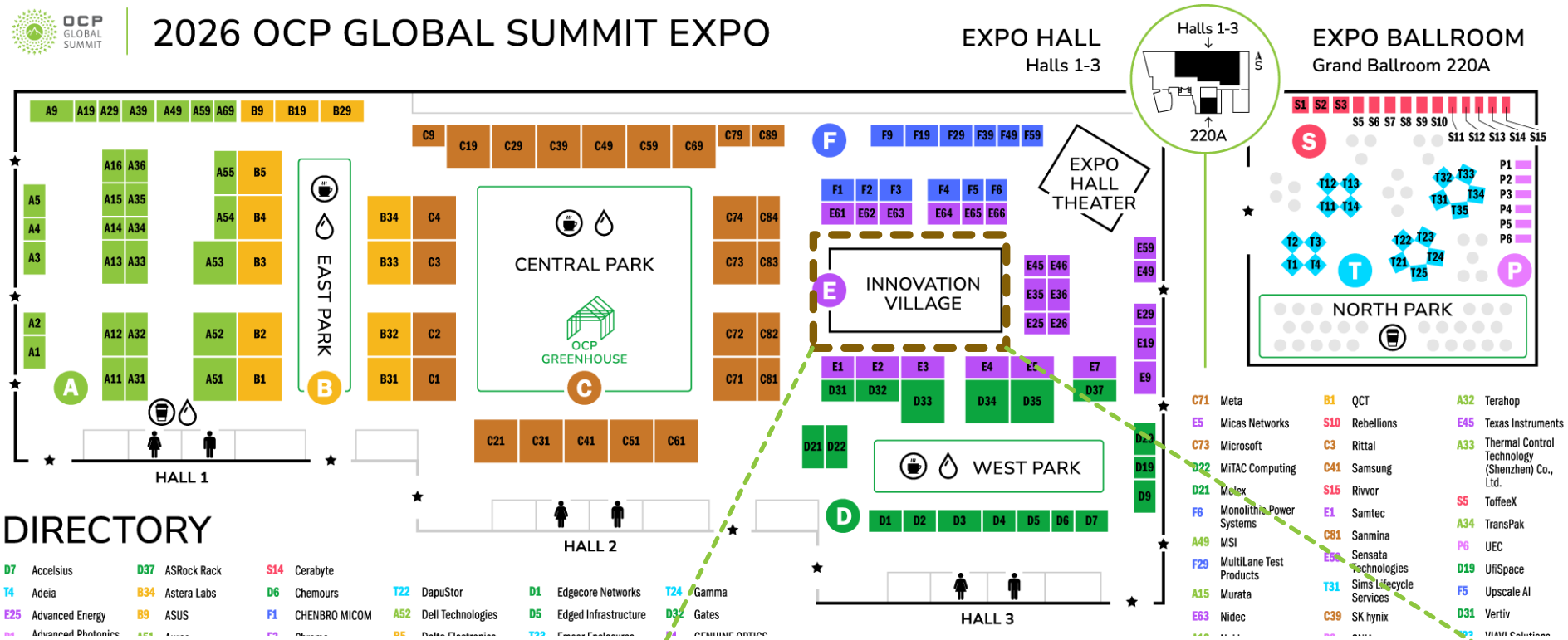
Exciting time: µVCSEL is the solution of choice for wide-and-parallel interconnect

Collaborate: We welcome partnerships with research labs and industry leaders

Hiring: We are growing across photonics, circuits, packaging, and systems

LightXcelerate — Photonics Empowering Next Computing Paradigm

2026 OCP GLOBAL SUMMIT EXPO

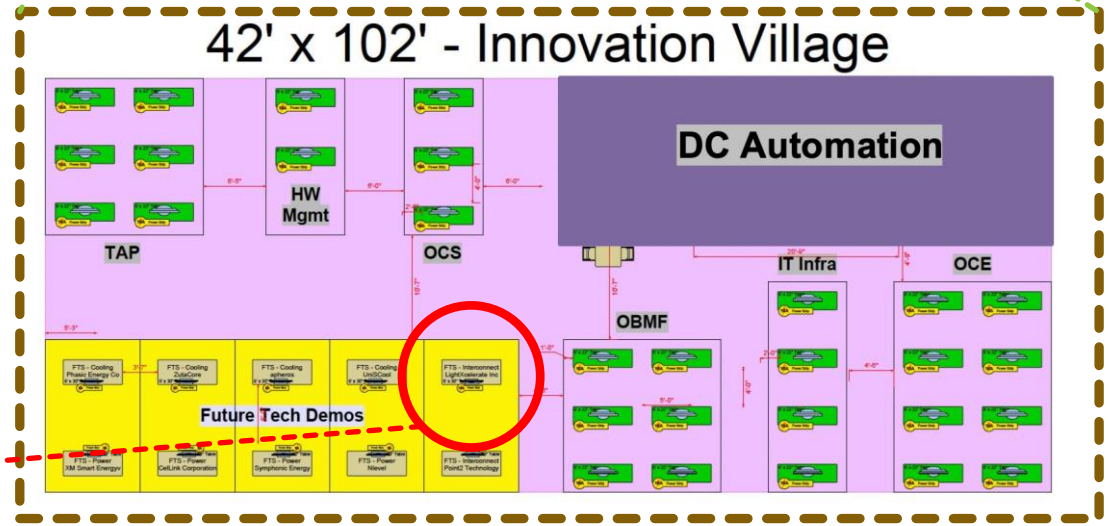


LightXcelerate Live Demo

Innovation Village, OCP Exhibition Hall

Oct 13 – 15, 2026

(Email to book a time with LXC team: manchen@lightxcelerate.com)



We are here

OPEN Compute Project®

OCP FUTURE TECHNOLOGIES SYMPOSIUM

October 14, 2026
San Jose, California

